

Study of Cybersickness Prediction in Real Time Using Eye Tracking Data

Shogo Shimada*

Tokyo Metropolitan University

Yasushi Ikei†

The University of Tokyo

Nobuyuki Nishiuchi‡

Tokyo Metropolitan University

Vibol Yem§

Tokyo Metropolitan University

ABSTRACT

Cybersickness seriously degrades users' experiences of virtual reality (VR). The level of cybersickness is commonly gauged through a simulator sickness questionnaire (SSQ) administered after the experience. However, for observing the user's health and evaluating the VR content/device, measuring the level of cybersickness in real time is essential. In this study, we examined the relationship between eye tracking data and sickness level, then predicted the sickness level using machine learning methods. Some characteristics of eye related indices significantly differed between the sickness and non-sickness groups. The machine learning methods predicted cybersickness in real time with an accuracy of approximately 70%.

Keywords: Cybersickness, VR sickness, real-time prediction, eye tracking, machine learning, virtual environments.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality;

1 INTRODUCTION

Many head-mounted displays (HMDs) with highly portable designs, such as the Oculus (Meta) Quest 2, have been popularized and VR technology is becoming more familiar to general users. However, VR experiences are known to cause discomforting symptoms such as eye strain, headache, sweating, disorientation, and nausea, which resemble those of motion sickness [1]. This side effect, commonly referred to as cybersickness or VR sickness, persists in the general use of VR devices. Thus, cybersickness is a major obstacle to technological development and methods that suppress or avoid cybersickness during VR immersion should be investigated in further studies. To check the effectiveness of countermeasures against cybersickness, a method for real-time detection of cybersickness is needed in advance.

In recent years, eye tracking data can now be measured by sensors embedded in the HMD, opening avenues for detecting cybersickness and predicting the sickness level from eye tracking data alone [2]–[4]. However, to our knowledge, no previous study has investigated the predictability of cybersickness occurrence using eye tracking data processed at high frequencies. In addition, in many studies, the sickness level induced by VR is subjectively evaluated through questionnaire responses. The SSQ [5] is widely applied for this purpose. Although the cybersickness level must be measured in real time to observe the user's health and to evaluate VR contents and devices, the SSQ allows only post-event evaluation, so cannot provide sufficient information during the VR immersion period.

Considering these backgrounds, in this study, the authors examined the association between specific eye related indices and the degree of cybersickness obtained during VR immersion, then applied them to a machine learning method to predict the occurrence of cybersickness in real time at a high frequency of 1-second intervals.

*e-mail: shogo-shimada@ed.tmu.ac.jp

†e-mail: ikei@vr.u-tokyo.ac.jp

‡e-mail: nnishiuc@tmu.ac.jp

§e-mail: yem@vr.sd.tmu.ac.jp



Figure 1: Screenshots of the three VR videos shown while collecting the eye tracking data of experimental participants: (a) car video, (b) roller coaster video, (c) relaxing video.

2 MEASUREMENT OF CYBERSICKNESS AND EYE TRACKING

2.1 Experimental Setup

23 participants (19 males and 4 females) were recruited. The study was approved by the Research Ethics Committee of Tokyo Metropolitan University (Approval number: H22-029). The VR headset used in this study was the HTC Vive Pro Eye. As cybersickness-inducing scenes, we selected a car video and a roller coaster video (Fig. 1 (a), (b)). In the other video (Fig. 1 (c), here after called the relaxing video), the user was “seated” beside a quiet lake, which was intended to induce a calm, relaxed sensation. No sickness-reducing technology was applied to these VR videos.

In this experiment, each participant sat on a chair and watched a combination of these prepared videos. Two conditions were prepared per participant. Participants watched one of the sickness videos (car or roller coaster contents) per condition. After watching one set of scenes, the participants were allowed to rest until their cybersickness symptoms recovered. The sickness video was preceded and followed by the relaxing video. Each sickness video was approximately five minutes long and the relaxing video was one minute long. During the experience, the participants subjectively indicated their sickness level by placing their thumb at the appropriate position on the trackpad of the HTC VIVE controller (2018). The trackpad is equipped with a sensor which allows continuous recording values by simply placing a finger on it. Similarly to the SSQ, developed as a subjective evaluation measure of simulator sickness, we evaluated four levels of cybersickness: None, Slight, Moderate, and Severe.

2.2 Data Processing

We collected 11 types of eye tracking data: normalized pupil positions of both eyes (x, y), pupil diameters of both eyes [mm], normalized gaze point deviation from the center point, angular velocity of the eye gaze at 0.02-s intervals [deg/sec], and gaze origin position [mm] (x, y, z). All of these data were recorded in the right-handed coordinate system at 50 Hz and were preprocessed as follows:

- Nine types of eye tracking data (normalized pupil positions and pupil diameters of both eyes and the gaze origin position) of each participant obtained during the sickness videos were divided by their corresponding average value over the last 30 seconds of the relaxing video presented before the sickness video. After this normalization process, the differences caused by individual characteristics were suppressed and the eye tracking data obtained while watching the relaxing video could be compared with those obtained while watching the sickness (car and roller coaster) video.
- The dataset was converted into a new dataset with one-second intervals by calculating the mean over each of the 50 inter-

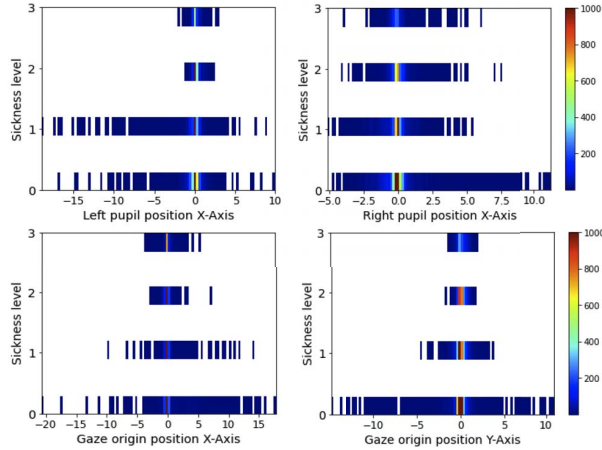


Figure 2: Distributions of sickness levels (0: None, 1: Slight, 2: Moderate, 3: Severe) in various eye tracking data

vals. To account for the reaction-time delay in cybersickness evaluations, a reaction time of 0.5 s was selected based on the experimental results of Nalivaiko et al [6]. This step suppressed the influence of noise generated by extracting the eye tracking data at high frequency (50 Hz).

3 RESULTS AND DISCUSSION

3.1 Eye Related Indices and Cybersickness

We examined the relationship between the obtained eye tracking data and sickness level. The following results were obtained by combining and analyzing the eye tracking data obtained while watching the car and roller coaster videos. Fig.2 shows the distributions of some eye tracking data at different sickness levels (0: None, 1: Slight, 2: Moderate, 3: Severe). The characteristics of several eye related indices differed between the “None and Slight Sickness” groups (hereafter called the non-sickness group) and the “Moderate and Severe Sickness” group (hereafter called the sickness group). To verify the statistical significance of these differences, we applied the Kolmogorov–Smirnov test and showed $p < 0.01$ on the distributions of these eye tracking data in the sickness and non-sickness groups. The x -axis of pupil position, the x - and y -axis of the gaze origin position showed significantly different probability distributions in the sickness and non-sickness groups. To our knowledge, the x - and y -axis of gaze origin position have not been previously associated with sickness level, but both features might be useful predictors of sickness level depend on the data-processing approach.

3.2 Cybersickness Prediction

As the relationship between eye tracking data and cybersickness level is complicated and the level of sickness is not easily predicted from raw data observation, we adopted a machine learning approach. The possibility of predicting the sickness level in each dataset was evaluated at 1.0-s intervals through binary classification tasks (sickness versus non-sickness groups). As the machine learning model is intended for real applications in future. Accordingly, the cross-validation of the datasets obtained by watching the car and roller coaster movies was repeated using one dataset as the training data and the other as the test data. The mean accuracy, precision, and recall results of this test are shown in Table 1. Moreover, the machine learning models achieved extremely high recall results in the non-sickness group, exceeding 90% in one case (the actual result depended on the machine learning algorithm). In stark contrast, the recall scores were low in the sickness group. This result indicates that when the sickness group alone is to be detected among the obtained dataset, that group will be detected at low rates. The

Table 1: Accuracy, precision, and recall scores in cross-validation using the car or roller coaster video dataset for training and the other dataset for testing.

ML Algorithms	Accuracy	Precision		Recall	
		non-sickness	sickness	non-sickness	sickness
Random Forest	0.6863	0.7343	0.4072	0.8782	0.2085
LightGBM	0.6809	0.7372	0.4032	0.8588	0.2376
SVM(rbf)	0.6985	0.7415	0.4490	0.8864	0.2376
SVM(poly)	0.7051	0.7350	0.4614	0.9175	0.1760
MLP(ReLU)	0.6739	0.7421	0.4009	0.8320	0.2799
MLP(tanh)	0.6780	0.7486	0.4163	0.8262	0.3088

performance results of the machine learning model further suggest that cybersickness might be difficult to detect because many characteristics of the eye tracking data in the sickness group are overlapped by those in the non-sickness group. The prediction performance of cybersickness using eye tracking data should be improved in further studies.

4 CONCLUSION

In the present study, validation was limited to the data obtained from participants watching the car and roller coaster videos. Another future task is verifying whether the method performs similarly on a dataset of participants watching other videos with slowly changing visibility. For instance, a ship video will embed different characteristics from those of the car and roller coaster videos. Moreover, we processed the data at each second, which limits the speed of the cybersickness prediction in real time (although our prediction ran faster than previous studies). To improve both the speed and performance of cybersickness detection, we suggest a time-series analysis with reference to past eye tracking data, such as fluctuations of values within a certain interval. Although the current results are imperfect, our findings from eye tracking data may provide clues for cybersickness detection.

ACKNOWLEDGMENTS

This research was partially funded by JSPS KAKENHI grant number JP20K12511 and JP18H04118, and local-5G project of Tokyo Metropolitan University.

REFERENCES

- [1] Joseph. J and LaViola. Jr. A discussion of cybersickness in virtual environments. SIGCHI Bull, 32(1), pages 47-56, 2000.
- [2] Wibirama. S, Santosa. P. I, Widyarani. P, et al. Physical discomfort and eye movements during arbitrary and optical flow-like motions in stereo 3D contents. Virtual Reality 24, pages 39-51, 2020.
- [3] Lopes. P, Tian. N, and Boulic. R. Eye thought you were sick! Exploring eye behaviors for cybersickness detection in VR. In Proceedings of the 13th ACM SIGGRAPH Conference on Motion, Interaction and Games (MIG '20). Association for Computing Machinery, New York, NY, USA, Article 3, pages 1-10, 2020.
- [4] Chang. E, Kim. H. T, and Yoo. B. Predicting cybersickness based on user's gaze behaviors in HMD-based virtual reality. Journal of Computational Design and Engineering, 8(2), pages 728-739, 2021.
- [5] Kennedy. R. S, Lane N. E, Berbaum. K. S, and Lilienthal M. G. Simulator sickness questionnaire: An enhanced method for quantifying simulator sickness. The International Journal of Aviation Psychology, 3(3), pages 203-220, 1993.
- [6] Nalivaiko. E, Davis. S. L, Blackmore. K. L, Vakulin. A, and Nesbitt. K. V. Cybersickness provoked by head-mounted display affects cutaneous vascular tone, heart rate and reaction time. Physiology and Behavior 151, pages 583-590, 2015.